MAVERICK MINI HACKATHON

5 AI Business Projects

Five domains · five briefs · five answer keys

Everything a team needs to build, and everything a judge needs to grade.

Format	Teams of 3–4 • 4 hours build + 3-min pitches • each team gets ONE project (assigned or drawn by lots)
Stack	$0 — no Azure. Python + scikit-learn + MLflow + Streamlit. Every dataset is free (Kaggle / UCI) and laptop-sized.
Skills tested	Exactly the bootcamp arc: framing → EDA → leakage-free modelling → the right metric → an actionable output → a business pitch.
This pack	Part A = organizer guide + common rules & rubric. Part B = the five project briefs (share each with its team). Part C = judges' answer keys (organizers ONLY).

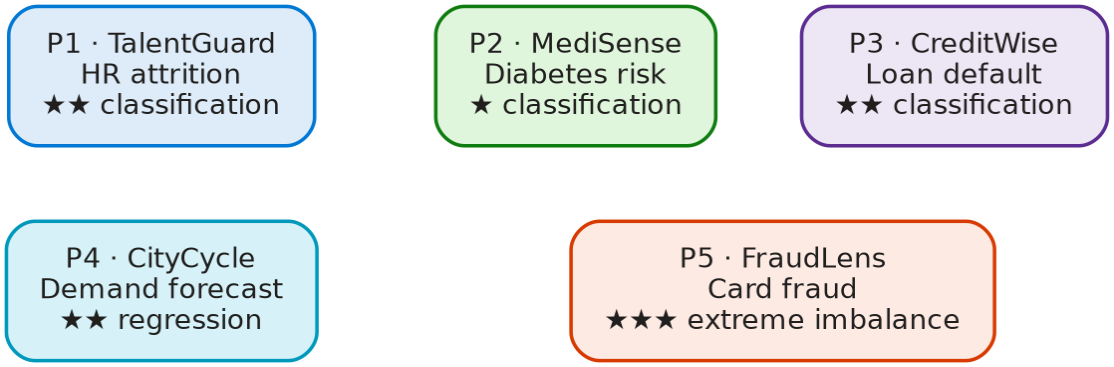


Figure: Five domains, one skill set — difficulty stars help you match projects to teams.

PART A — Organizer guide (rules, rubric, logistics)

A1. Why five projects

Giving each team a different domain kills copy-paste between teams, makes the pitch round genuinely interesting, and lets you match difficulty to team strength: ★ **MediSense** is the gentlest, ★★ **TalentGuard / CreditWise / CityCycle** are the core tier, and ★★★ **FraudLens** is for a confident team. All five test the same bootcamp skills, so one rubric grades them all.

- **Assigning:** draw lots for fairness, or assign by team strength. With 10+ teams, repeat projects but never give two neighbouring teams the same one.
- **Regression note:** CityCycle (P4) is a regression problem — its 'right metric' is MAE/RMSE vs a naive baseline instead of recall/F1. The rubric below already accounts for this.

A2. Common rules

- Teams of 3–4. Any Python libraries; any free tool. No paid services; no secrets in repos.
- AI assistants (Copilot/ChatGPT/Claude) are ALLOWED — but you must understand and be able to explain every line. Judges will ask.
- Submit by deadline: GitHub repo link + your output artifact (watch list / forecast / alert queue) + max 3 slides or an app URL.
- Pitches: 3 minutes + 2 minutes of judge questions; every member speaks.
- Copied kernels without understanding = disqualification. It shows in 30 seconds of questioning.

A3. Common scoring rubric (100 points + bonus, all projects)

Task	Points	What judges look for
T1 — Business framing	10	The decision being improved + a business success metric (money/capacity), not a model metric.
T2 — EDA & insights	20	3+ evidence-backed findings; the project's hidden data traps found and handled (every project has at least one — see the answer keys).
T3 — The model	30	A baseline + a stronger model; the RIGHT metric for the problem (recall/F1/PR-AUC for classification, MAE/RMSE vs naive baseline for regression); threshold/operating point tied to a stated capacity; no leakage.
T4 — Actionable output	20	The project-specific artifact (defined in each brief): a ranked list / forecast / alert queue WITH reasons and a recommended action.
T5 — The pitch	20	3-minute story ending in business value; one live scoring/prediction on stage.

BONUS (up to +15, any project)

• +5 MLflow experiment tracking with a run comparison.

• +5 Free public deployment (Streamlit Community Cloud).

• +5 An 'agent' touch: auto-draft the action (retention email / follow-up letter / rebalancing order / fraud-review note) marked PENDING_APPROVAL — rule-based is fine.

A4. Timeline (4 hours)

Time	Do this
0:00–0:20	Read your brief, download data, set up repo, split roles.
0:20–1:10	EDA sprint — target balance/shape, key drivers, find the traps.
1:10–2:20	Model sprint — leakage-free Pipeline, baseline, stronger model, right metric, operating point.
2:20–3:10	Output sprint — the actionable artifact + reasons + recommendations; (bonus) app/MLflow/deploy.
3:10–3:40	Pitch prep — 3 slides, rehearse, pick your live example.
3:40–4:00+	Pitches + judging + winners.

PART B — The five project briefs

Print or send each brief to its team. Each brief is self-contained: the business story, the data, the required output, and the traps warning (without revealing the traps).

PROJECT 1 — TalentGuard (HR / People Analytics) ★★

Client	TechNova Solutions, a 12,000-person IT services firm losing 16% of staff a year; every exit costs 6–9 months of salary.
The ask	"Tell me who is likely to resign next cycle, why, and what to do — while there's time to act." HR can run retention talks with 50 employees per cycle.
Dataset	IBM HR Analytics Employee Attrition (Kaggle: ibm-hr-analytics-attrition-dataset) — 1,470 employees, 35 columns, target Attrition (Yes/No).
Required output (T4)	A ranked TOP-50 watch list with each employee's top-3 risk reasons + a concrete retention recommendation for the top 10.

Watch out: this dataset contains an ID column and several useless columns that can hurt your model — finding and handling them is part of the score. And think hard about which metric matters when only ~1 in 6 employees actually leaves.

PROJECT 2 — MediSense (Healthcare) ★

Client	A community health clinic screening patients for type-2 diabetes risk. Untreated diabetes is costly and dangerous; early lifestyle intervention works.
The ask	"Give our nurses a risk score so they know which patients to call back first." The follow-up team can call 100 patients per month.
Dataset	Pima Indians Diabetes (Kaggle: uciml/pima-indians-diabetes-database) — 768 patients, 8 clinical features (glucose, BMI, age, blood pressure…), target Outcome (0/1).
Required output (T4)	A ranked TOP-100 call-back list with each patient's top risk factors + a one-line nurse note per top-10 patient (what to discuss on the call).

Watch out: look very carefully at the minimum values of the clinical columns before trusting any statistics — something about them is medically impossible. With only 768 rows, also think about how to make your evaluation stable. Finally: a false negative and a false positive do NOT cost the same in healthcare — say which matters more and why.

PROJECT 3 — CreditWise (Banking) ★★

Client	A consumer credit-card issuer in a competitive market. Defaults eat margins; over-tight limits push good customers away.
The ask	"Predict which customers will default on next month's payment so we can act early — proactive limit reviews, payment plans, or outreach." The risk team can review 500 accounts per cycle.
Dataset	Default of Credit Card Clients (UCI, also on Kaggle: 'default of credit card

	clients') — 30,000 customers, 23 features (credit limit, past payment status, bill & payment amounts, demographics), target = default next month.
Required output (T4)	A ranked TOP-500 risk review queue with reasons + a recommended action tier (call / limit review / payment plan) for the top 20.

Watch out: read the column documentation — the repayment-status codes are cryptic, some category columns contain undocumented values, and one column must never be a feature. There's also an ethics question hiding in the demographic columns: decide what you use, and defend it in the pitch.

PROJECT 4 — CityCycle (Mobility / Operations) ★★ — REGRESSION

Client	A city bike-share operator. Empty docks at 8am and full docks at 6pm mean lost rides and angry riders; trucks rebalance bikes overnight and during the day.
The ask	"Forecast hourly demand so we can plan rebalancing and staffing." Ops needs the forecast and the WHY (weather? hour? season?) to trust it.
Dataset	Bike Sharing Dataset (UCI / Kaggle: bike-sharing-dataset, use hour.csv) — 17,379 hourly records over 2 years: weather, temperature, season, working day, and the rental count.
Required output (T4)	An hourly demand forecast for a held-out period, an honest error number (MAE/RMSE) compared to a naive baseline, + a one-page 'ops guidance': the 3 biggest demand drivers and what ops should do about them.

Watch out: two of the columns will give you a perfect — and perfectly useless — model. Find them. And think about HOW you split time-series data for testing: one obvious choice quietly cheats.

PROJECT 5 — FraudLens (Fintech / Payments) ★★★

Client	A card issuer processing ~300k transactions in two days. Fraud is rare but each miss costs real money and customer trust; each false alarm annoys a customer and burns analyst time.
The ask	"Build the triage brain: which transactions should our fraud analysts review first?" The review team can handle 200 alerts per day."
Dataset	Credit Card Fraud Detection (Kaggle: mlg-ulb/creditcardfraud) — 284,807 transactions, anonymized PCA features V1–V28 + Time + Amount, target Class (fraud = 1).
Required output (T4)	A ranked TOP-200 daily alert queue with a precision/recall statement at that cutoff ("of 200 alerts, ~X are real fraud; we catch ~Y% of all fraud") + a one-line analyst note format.

Watch out: the target is astonishingly rare — your first 'amazing' metric will be a lie; pick the metric that survives that. If you use any resampling trick, be extremely careful WHERE in your workflow you apply it. And ROC-AUC will flatter you here; there's a better curve for this problem.

PART C — Judges' answer keys (organizers ONLY)

> **CONFIDENTIAL**
> Do not share until after judging. Numbers are expected ranges on the standard public datasets; exact values vary with seed and split — grade the reasoning, not the third decimal. Every project's hidden traps are listed; finding them is a big part of T2/T3 marks.

KEY — P1 TalentGuard (HR attrition)

- **Facts:** 1,470 × 35; no missing values; Attrition=Yes for 237 ≈ 16.1% (all-No accuracy ≈ 84% — the trap).
- **Hidden traps:** `EmployeeNumber` (ID — drop); `EmployeeCount`=1, `StandardHours`=80, `Over18`=Y (constants — drop).
- **Expected EDA:** OverTime ≈ 30% attrition vs ≈ 10% (≈3×); Sales Representative ≈ 40% (highest role); low income, low tenure/age, low satisfaction, frequent travel all elevate risk.
- **Expected model:** logistic (class_weight=balanced) ROC-AUC ≈ 0.80–0.85; boosting similar — the baseline is competitive and honest teams say so. AUC > 0.95 = leakage red flag (ask what columns).
- **Operating point:** top-50 by probability (HR capacity); expect precision@50 well above the 16% base rate.
- **Judge Q:** "Why not accuracy?" / "What did you fit on train only?" / "Is OverTime causal?" (predictive, not proven causal — recommend an intervention and measure).

KEY — P2 MediSense (diabetes risk)

- **Facts:** 768 × 9; Outcome=1 for 268 ≈ 34.9% — mildly imbalanced.
- **Hidden trap (the big one):** zeros in Glucose, BloodPressure, SkinThickness, Insulin, and BMI are **physiologically impossible — hidden missing values** (Insulin ≈ 49% zeros, SkinThickness ≈ 30%). Full T2 marks require treating zeros as missing (impute median + was-missing flag), not as real values.
- **Expected EDA:** Glucose is the dominant driver, then BMI, Age, and DiabetesPedigreeFunction; pregnancies correlate with age.
- **Expected model:** ROC-AUC ≈ 0.80–0.84 (logistic or boosting). With 768 rows, cross-validation (or repeated splits) is the mature move — single-split scores swing ±0.05.
- **Cost asymmetry answer:** a false negative (missed diabetic) is worse than a false positive (an unnecessary call) → favour recall; the top-100 list at ~35% base rate should be majority true positives.
- **Judge Q:** "What did the zeros mean?" / "Why cross-validate here?" / "Which error is worse and how did that change your threshold?"

KEY — P3 CreditWise (credit default)

- **Facts:** 30,000 × 24; default rate ≈ 22.1%; no true missing values.

- **Hidden traps:** `ID` column (drop); PAY_* repayment codes are cryptic (-2/-1/0 = variants of ‘paid/no balance’, 1+ = months delayed); EDUCATION contains undocumented 0/5/6 and MARRIAGE contains 0 (collapse into ‘other’).
- **Ethics answer (award points for engaging):** SEX/AGE/MARRIAGE as features raise fairness concerns for credit decisions — strong teams either drop demographics or show performance barely changes without them (it barely does: behaviour features dominate).
- **Expected EDA:** PAY_0 (most recent repayment status) is by far the strongest signal, then PAY_2/PAY_3; lower LIMIT_BAL correlates with default; bill/payment ratios add a little.
- **Expected model:** this dataset tops out around **ROC-AUC ≈ 0.76–0.78** with boosting; logistic ≈ 0.72–0.75. A team reporting 0.85+ almost certainly leaked or mis-split — probe hard.
- **Operating point:** top-500 of 30k (≈ top 1.7% by risk if scored on all; or scaled to their test set) with precision far above the 22% base rate.
- **Judge Q:** “What does PAY_0 = -1 mean?” / “Would you use gender in a credit model?” / “Why is 0.78 AUC still valuable?” (at 22% base rate, doubling precision in the review queue is real money).

KEY — P4 CityCycle (demand forecasting — regression)

- **Facts:** hour.csv = 17,379 rows over 2011–2012; target `cnt` = hourly rentals.
- **Hidden trap #1 (leakage):** `casual + registered = cnt` — using either as a feature gives a near-perfect, useless model. Must be dropped. A suspiciously perfect R^2 (>0.99) = they used them.
- **Hidden trap #2 (time split):** a random split lets the model ‘see the future’ (adjacent hours in train and test). The mature answer is a **temporal split** (e.g., train on 2011–mid-2012, test on the last months). Random-split scores will look better; award marks for knowing why that’s cheating.
- **Expected EDA:** hour-of-day is the dominant pattern (twin commute peaks 8am/5–6pm on working days, single midday hump on weekends); temperature strongly positive (flattening when very hot); season/weather matter; year trend (2012 > 2011 — growth).
- **Expected model:** naive baselines first (predict the mean ≈ MAE ~140; or same-hour-last-week). Gradient boosting / random forest with hour, temp, season, workingday, weather typically lands **R^2 ≈ 0.85–0.94** on a fair split, MAE roughly in the 30–60 rentals range — accept a wide band, demand the naive-baseline comparison.
- **Ops guidance answer:** pre-position bikes before commute peaks on working days; weather-triggered staffing; treat weekends as a different demand regime.
- **Judge Q:** “Why did you drop casual/registered?” / “How did you split, and why not randomly?” / “What’s your error in bikes-per-hour and is that usable by ops?”

KEY — P5 FraudLens (card fraud — extreme imbalance)

- **Facts:** 284,807 transactions; 492 frauds = **0.172%**; features V1–V28 are PCA components + Time + Amount.
- **The accuracy lie:** all-negative scores 99.83% accuracy and catches zero fraud — the whole project is built to expose this. Headline metric must be **PR-AUC / precision-recall**

at a cutoff, not accuracy; ROC-AUC (≈ 0.95+) is technically high but flattering at this imbalance.

- **Hidden trap (resampling leakage):** applying SMOTE/oversampling **before** the split copies near-duplicates of frauds into both train and test → inflated scores. Correct: split (stratified) first; resample or class-weight the training fold only — or skip resampling and just use class_weight + threshold.
- **Expected model:** logistic (class_weight) PR-AUC ≈ 0.70–0.78; boosting ≈ 0.78–0.85. On a 25% stratified test (~123 frauds), a top-200 alert queue → expect roughly 60–100+ true frauds in it (precision ≈ 0.3–0.5+, recall ≈ 0.5–0.8) — accept ranges; demand the precision/recall statement, not a single number.
- **Expected EDA:** V14, V17, V12, V10 (and Amount patterns) separate classes most; fraud amounts skew smaller; Time has weak day-cycle structure. Scale Amount; V's are already scaled.
- **Judge Q:** "Why is 99.8% accuracy meaningless here?" / "Where exactly did you apply SMOTE and why?" / "At 200 alerts/day, what do precision and recall become, in plain words?"

Scoring bands & tie-breakers (all projects)

- **Winning band (85+):** traps found, leakage-free pipeline, right metric with a capacity-tied operating point, an artifact the client could use tomorrow, business-first pitch.
- **Solid (65–84):** working model + right metric, weaker artifact or a metric-led pitch.
- **Needs work (<65):** accuracy-led evaluation, leakage, or output the client can't act on.
- **Universal red flags:** too-good scores (P1 >0.95 AUC, P3 >0.85 AUC, P4 >0.99 R^2, P5 judged on accuracy), ID columns as features, no split-before-fit, code the team can't explain.
- **Tie-breakers (announce up front):** (1) quality of the actionable artifact, (2) trap-handling, (3) pitch clarity.